

Shane Holmes

Computer Engineer

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Summary

Computer Engineer with experience in embedded systems, robotics, and software development. Passionate about building things from first principles — from low-level embedded firmware to machine learning models and 3D rendering engines.

Education

Northeastern University, Master of Science in Computer Science – Boston, MA Jan 2024 – Jan 2026

- Expected graduation May 2026

Florida Polytechnic University, Bachelor of Science in Computer Engineering – Lakeland, FL Jan 2019 – Jan 2023

- Graduated Magna Cum Laude

University of Toronto, Exchange Student in Academic Exchange with Fulbright – Toronto, ON Sept 2022 – Jan 2023

Experience

Computer / Robotics / Embedded Engineer, Overhead Intelligence – Lake Wales, FL Oct 2023 – May 2024

- Worked on the electrical and computer systems of a battery-powered UAV
- Designed and implemented LiDAR-based computation and map synthesis, capable of mapping 1.2 km² in 20 minutes
- Collaborated with a multidisciplinary team of engineers and participated in in-field deployment
- Technologies: Raspberry Pi, C++/C, Python

Computer / Robotics / Software Engineer, SS Innovations (Formerly AVRA Medical Robotics) – Orlando, FL May 2022 – Sept 2022

- Worked on the computer vision sensor system for a surgical robot arm to identify facial landmarks
- Achieved accurate simulation test results for surgical motion and object identification
- Technologies: Python, OpenCV, Git

Research Assistant / Robotics & Software Engineer, Advanced Mobility Institute – Lake Wales, FL Jan 2021 – Jan 2022

- Developed a fleet of small robot vehicles to autonomously navigate and coordinate in an environment
- Built electrical motor interface systems and an ArUco sensor system to identify vehicles
- Achieved vehicle convoy movement and dynamic obstacle avoidance of objects greater than 3 inches consistently
- Technologies: Python, C++/C, OpenCV

Projects

Mock Rover — FSI Capstone Project Sept 2022 – May 2023

Reverse-engineered and rebuilt a non-functional mock rover for the NASA Space Grant Consortium program through Florida Space Institute.

- Led technical integration of embedded control systems designing a Raspberry Pi-based architecture coordinating four AC motors, stereoscopic camera feed, and wireless command interface
- Achieved ~100ft wireless range, 300lb payload capacity, and 5mph traverse speed
- Technologies: Python, C++, Raspberry Pi, Arduino Uno, motor control systems, stereoscopic cameras, 3D printing

Hand Robot — Commissioned Project Nov 2020 – Feb 2021

Designed and fabricated a fully articulated robotic hand with individual finger control and accompanying arm.

- Iteratively designed hand and arm in CAD with custom 3D printed components
- Engineered servo control circuits on Arduino platform for individual finger articulation
- Developed desktop application in C++ for real-time articulation control
- Technologies: Arduino, C++, CAD, 3D printing, servo control systems

3D CPU Software Rasterizer

Dec 2025 – present

Built a 3D software rendering engine from first principles in C++ with no GPU acceleration or rendering libraries.

- Full graphics pipeline — model space through screen space — implemented manually using 4x4 matrices
- Triangle rasterization using edge functions and barycentric coordinates
- Scene graph with hierarchical parent-child entity system and automatic world matrix propagation
- Technologies: C++, CMake, Eigen, SFML, CppUnitLite

Latent Explorer

Apr 2026 – present

End-to-end tool for training and interactively visualizing the latent space of image datasets in the browser.

- Trained a convolutional decoder on ~202k face images with PCA post-processing to surface principal axes of variation
- Exported model to ONNX and built a pure-JavaScript browser interface running inference with no backend
- Technologies: PyTorch, ONNX Runtime Web, scikit-learn, HDF5, Vanilla JS

Motor Babbling — RL for Inverse Kinematics

Jan 2025 – Apr 2025

Implemented DQN, DDPG, and SAC from scratch to teach a simulated planar robot arm to reach arbitrary targets in MuJoCo — no analytical IK solver.

- Compared value-based (DQN), deterministic actor-critic (DDPG), and stochastic actor-critic (SAC) against the same environment; SAC achieved ~80% success at 10cm threshold on 2-DOF random targets
- Built the MuJoCo environment from scratch with a configurable number of links and a shared BaseAgent interface for clean algorithm comparison
- Technologies: Python, PyTorch, MuJoCo, Gymnasium, NumPy, Matplotlib

Relevant Coursework

Mathematics: Probability and Statistics, Differential Equations, Linear Algebra, Discrete Mathematics, Analytical Geometry and Calculus (1, 2, 3)

Computer Engineering: Physics (1, 2), Computer Architecture and Organization, Microcomputers, Digital Logic Design, Circuits (1, 2), Digital Electronics, Computer Systems

Software Engineering: Operating Systems Concepts, Data structures and Algorithms, Object Oriented Programming, Computer Programming, Computer Networks, Secure Software engineering, Programming Design Paradigms, Database Systems, Foundations in Software Engineering

Robotics Systems: Autonomous Robotic Systems, Digital Signal Processing, Systems and Signals, Kinematics and control of Robotic Systems, Artificial Intelligence (1, 2), Machine Learning (1, 2), Computer Vision

Awards

Fulbright Canada Killam Fellowship: Fulbright Canada · 2022 · Competitive fellowship awarded to outstanding students for academic exchange between Canada and the United States.

Magna Cum Laude: Florida Polytechnic University · 2023-05 · Graduated with high academic distinction.

IEEE HKN Member: IEEE Eta Kappa Nu · 2023 · Member of the IEEE honor society chapter recognizing excellence in electrical and computer engineering.

President's List: Florida Polytechnic University · 2023

Bright Futures Scholarship: State of Florida · 2019

Florida Polytechnic University Institutional Scholarship: Florida Polytechnic University · 2019

Skills

Programming Languages: Python, C++, C, Java

Python Libraries: NumPy, Pandas, Matplotlib, OpenCV, Threading, PyTorch, TensorFlow, Keras, Scikit-learn

Embedded & Hardware: Raspberry Pi, Arduino, ESP32, STM32, Odroid, AVR, Linux, Embedded Linux

Robotics & Autonomy: Computer Vision, LiDAR, IMU, SLAM, GPS, UAV

Software Development: Git, OOP, Software Design, Software Testing, Software Documentation, Embedded Systems

Technical Areas: Machine Learning, Artificial Intelligence, Computer Vision, Digital Signal Processing, Robotics, Hardware, Software

Collaborative Tools: Jira, Confluence, Slack, Microsoft Teams, Zoom, Visual Studio

Languages

English: Native speaker